Worksheet 8.7

Newton's Third Law

Student: .... Class: ....

1. State Newton's Third Law of Motion.

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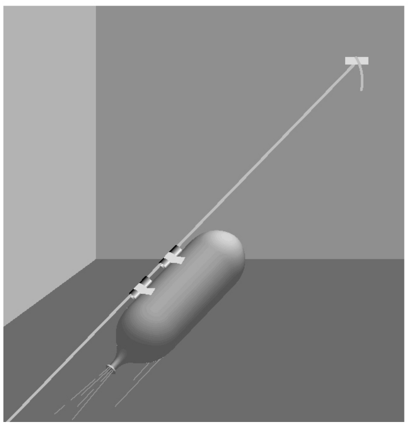
2. The following diagram shows a balloon rocket built by students to investigate motion. The balloon is mounted on a horizontal string using a length of drinking straw and masking tape.



The inflated balloon is then released.

(a) Where along the string will the balloon be moving at the greatest speed?

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(b) Use Newton's Third Law to explain why the balloon is propelled forward along the string.

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3. The photograph below shows a person pushing against a wall.

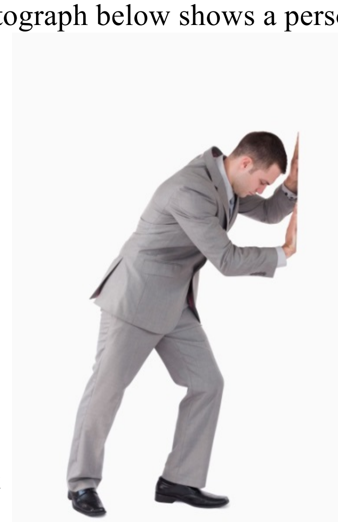

Photo: Wavebreak Media Ltd / 123RF

The man is not moving even though he is pushing forward against the wall.

(a) What is the net force acting on the man?

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(b) If the man pushes with a force of 200 N on the wall, what is the force that the wall exerts back on him?

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(c) The man is pushing back on the floor with his feet with a force 200 N. With what force does the floor push back on the man?

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(d) Explain what would happen if he tried to do this on a very slippery floor.

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